

The world as a playground: Ninja and parkour training

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Introduction

A welcome side effect of the increased interest in targeting the connective tissue by means of physical activity is the comeback of traditional exercises with swinging and bouncing movements seen in German gymnastics in the 19th century. Reviewing historical documents, the reappearance of different exercise modes uncovers that only the sociocultural frame has changed: the new is only the reinvention of the forgotten. In order to understand a big picture we have to revisit the term “sport” as a complex sociocultural construct (Heinemann, 2007).

Over the decades and centuries, there have been substantial changes in social interaction of our bodies that are strongly embedded into the sociocultural background of the society. Many movement cultures have their roots in the preparation of warriors and religious rituals. The development and differentiation of modern sports in the 19th century replaced the ancient movement cultures with new intentions, like health promotion, alongside social, political and economic functions (Scheid and Prohl, 2012).

For example, the theme of the ninja warrior currently is used in an international sports entertainment television show, in which 100 competitors attempt to complete a four-stage obstacle course. The ninja warrior appearance in TV sports entertainment would fit into the part of the economic and entertainment orientation

of our society. Another rediscovery of the ninja warrior theme can be identified in the parkour movement culture. This urban subculture is characterized by the goal of getting from point A to B in the most efficient and straightforward way by using only the body's skills. In this context, overcoming any obstacle is a main characteristic of parkour, which shares common ground with the characteristic of the ninja warrior theme. Both movement cultures have in common that the connective tissue is used and trained to maximize movement efficiency.

In contrast to the economic orientation of the sports entertainment television show and other obstacle race mass events, parkour serves as a non-commercial countermovement to our profit-oriented highly technical society, one leading us to bodily dissociation and disembodiment, independent of physical abilities. It is also worth mentioning that other modern movement cultures (e.g. hardcore bodybuilding) serve to replace absent physical labor (Klaeber, 2014). The perspective of historical sport science shows that ninja training overlaps with sports such as climbing, and street-based subcultures like calisthenics, with their classical branch of gymnastics, alongside Georges Hébert's *Méthode Naturelle*.

The aim of this chapter is to use a bottom up approach to identify fascia friendly movement patterns in ninja style training and to illustrate how the involvement of connective tissues supports movement efficiency.

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Basic considerations

Ninja style training or parkour lead to a reinterpretation of our environment. A gym is not necessary, you simply use what the environment provides. In contrast to standardized movements, such as those found in gymnastics, yoga, and pilates, ninja-style movements must adapt to the changing environment. Depending on the dimensions, surface and shape of obstacles, single movement phases are extremely variable. Thus, an exact standardization is not possible and there is never just one and only way of overcoming an obstacle. Creativity plays a vital role, but the scope for creativity is strongly dependent on individual skill, strength, body awareness, perception of the environment, willingness to reflect on one's own actions, and an adequate estimation of one's own abilities. The fascination of effortlessness in overcoming obstacles is the result of deliberate practice. Besides prolonged mental and strength training, the internalization of movements plays an essential role to make movements unconsciously available in different situations. From a perspective of motor learning, the internalization of a movement aligns with using less strength, increased efficiency, fluidity and control, because of better coordinated motor programs (Hirtz et al., 2003). From a biomechanical point of view, if less force is used then more elastic energy storage must be used, as seen in Kenyan endurance runners (Sano et al., 2015) or good animal movers like kangaroos, which use their tendons to save energy and require less muscle work.

The following sections provide some basic cues on what you have to focus upon to become an efficient mover using the properties of fascia connective tissues. Before we start with practical analysis of some movement patterns, some general guidelines are important:

- **Safety:** Ninja training is not kamikaze training. Movement control is paramount. Never go beyond your abilities.
- **Sustainability:** Injury and long-term overload for joints and myofascial tissues should be avoided by using the approach of small steps. Sometimes a step back is a step forward. A movement accomplished once does not denote mastery. Practice movements on both sides to prevent imbalances. It pays off to train your weak or unnatural side to challenge your tensegral network and motor control in a new way.
- **No competition:** Our society and many sports educate us to become competitive at an early age. This ignores the individual foundations, learning and adaptation curves of each individual. The rejection of the competitive idea serves to prevent destructive behavior.
- **Efficiency:** The goal is to lower the investment of your resources to accomplish a movement task. Efficiency stands in contrast to effectiveness, where you invest all available resources to reach a goal.
- **Control:** Movement control is strongly related to safety. A controlled movement is fluid, smooth and silent.

Ninja movement patterns

Ninja movements can be classified as jumping, landing, falling, hanging, supporting, pulling up, rolling on the ground, swinging, turning, balancing, moving hand over hand, sliding, and quadrupedal movements. Together, these movements cover all possible human locomotion and, in ninja training, they are used on a regular basis. The following examples represent a small excerpt of the comprehensive movement

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repertoire in ninja parkour training. The selection focuses on light to medium skills that, to a certain extent, can be incorporated into a regular fascial fitness program.

Balancing

A pre-requisite for all ninja skills is balance. This is the foundation that permeates the whole spectrum of ninja movements. Different situations require very specific balance skills: to improve your balancing skill, many structures and techniques are beneficial. There are, however, differences compared with classical balancing skills, like those seen in rope artists in the circus. These artists usually turn their heels inwards. If you are balancing on rails or similar structures then this is not recommended because subsequent movements are restricted in the ankle joints. An increase in balancing speed can lead to increased balance in the short term but at the expense of control.

The physiology of balancing is based on the integration of inputs from the visual, vestibular and proprioceptive system. Looking at different people and their balancing strategies, it can be concluded that the relative contribution of these systems seems very individual. If someone has poor vision, it can be compensated to a certain

degree by relying more on using proprioception. If we keep in mind that fascia is one of our most important organs for proprioception (Findley and Schleip, 2007), then working on your balancing skills can be considered as working on fascia as a sensory organ (see Figure 35.1).

Jumping off

To overcome obstacles, a parkour ninja must be able to jump off and achieve a large jumping distance. In this case very basic biomechanics come into play. Jumping distance can be significantly increased by utilizing a countermovement. The assumed physiological mechanisms behind the performance enhancement using a countermovement are described in more detail in Chapter 25.

Warm up for an experiment, then do three jump-offs from a standing position. In the first attempt bend your knees only a little (Figure 35.2A), in the second attempt jump from a deep squat (Figure 35.2C), and in the third attempt from a half squat position (Figure 35.2B). You will notice that the medium position (Figure 35.2B) shows the best result in jumping distance. In practice you have to find your optimal interaction of acceleration path and joint angles. When it feels good, then it is right. This is also



Figure 35.1

Balancing on a handrail in bipedal or quadrupedal mode is a valuable exercise to develop proprioceptive perception.

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dependent on your individual anthropometrics of the lower limbs and muscle fiber composition. A fluid coordination of single movement phases and body parts leads to another increase in jumping distance, like a perfectly integrated arm swing. If the timing of the double arm swing is inappropriate, then you do not achieve an increase in jumping distance compared to a jump off without using a double arm swing.

Landing and falling

Falling means that the body is in a downward motion that ends with an impact. Do not jump before you have learned how to land. Depending on the task, a parkour ninja must be able to land silently in precisely any position they want. This is of great importance in the precision jump, where a very small landing surface must be hit

as accurately as possible without losing balance. The mechanical stress on the body for untrained individuals is very high in the precision jump (see Figure 35.3) because the landing is finished in a deep squat position on the balls of the feet. The sound intensity of the landing gives you a precise feedback of the landing quality.

A derivation for the purpose of fascial health is to perform barefoot precision jumps for short distances on different surfaces. Besides the mechanical properties, this form of exercise promotes proprioceptive feedback and a differentiated interaction with your body.

In order to prevent harm, the aim of controlled falling (e.g. after a depth jump) is the amortization of the energy of the accelerated body. Landing technique is of great importance for

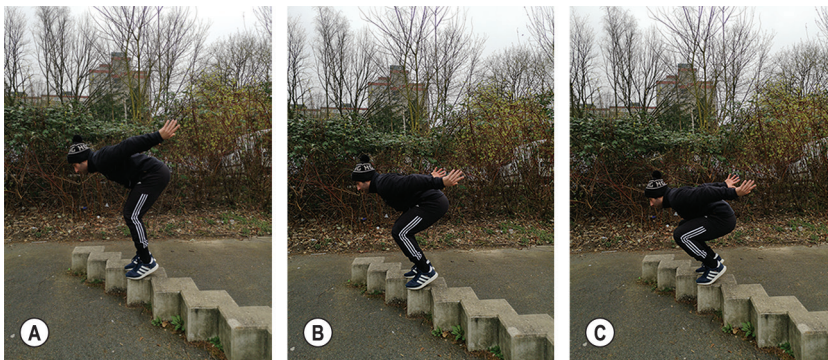


Figure 35.2

Jumping with different counter-movement lengths. For each individual there is an optimal angle that leads to the best jumping performance.

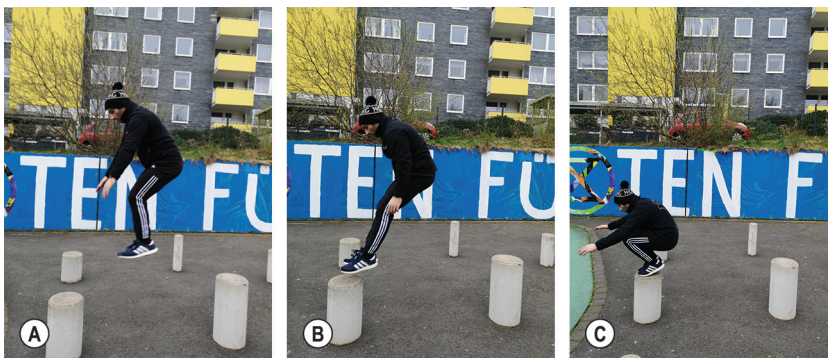


Figure 35.3

Precision jump on small landing surface with landing on the balls of foot.

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amortization of the downward motion into a forward motion. There are different unrolling and bounced landing techniques to master this movement task. For example, the impact of a bounced landing (see Figure 35.4) is cushioned by a shift of the body's center of gravity and distribution of the amortization on arms and legs. By using the reactive force (Chapter 25), the vertical energy can be transferred into a locomotion movement. Note that when using the elastic properties of connective tissues, the ground contact time has to be very short, so that you avoid contact with your heels. A long “sit down” position would prevent force distribution and lead to high peak forces in the joints due to unfavorable joint angles (Luksch, 2009).

Hanging

All hanging movements can be considered as fascia-friendly exercises, because gravity provides a strong stretch stimulus on the whole body in every hanging position. They also represent very basic movement patterns that serve as a foundation for a number of skills. These additional skills promote complex flexibility and the movement skill of a ninja. There is no correct way of hanging when we focus on health aspects of fascia. The infinite degrees of freedom are the basic principle here. Thus, you can repeatedly change body

angles to target the desired fascial tissue areas. For example, take a deep squat position and hang up on a balustrade (see Figure 35.5). While you keep tension in the upper body, you begin to bounce in this position. In a ninja parkour progression this is the introduction to the cat leap or arm jump, in which the ninja jumps from the run-up onto the wall.

Monkey bars

When swinging from bar to bar like a monkey, grip strength is the limiting factor to mastering this skill. There is a ninja secret that uses the principle of an inverted pendulum in deviation movements: if you bend your upper arms, you are cutting off body-wide force transmission, using unnecessary upper arm muscles, so you are wasting strength. Bar to bar movements you should consider as a one-handed swing instead. Watch a gibbon's elegant swinging from branch to branch; they hardly bend their elbows in horizontal deviation movements. The same is true for children in the playground. They are able to catch bar to bar, but as a result of their lack of upper arm strength, they are unable to bend their elbows. Nevertheless, in order to train fascia you definitely need strong muscles. This gives us a hint for the importance of muscular strength training as a pre-requisite in order to exercise fascia connective



Figure 35.4

Example of a bounced landing technique with amortization into a forward locomotion.

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Figure 35.5

Examples of hanging exercises.

tissues properly. The muscles involved work in a quasi-isometric contraction mode while using the ape-style swing from bar to bar. Interestingly, the muscle's isometric contraction mode can also be observed in other movement patterns, where the elastic properties of fascia predominantly come into play (Chapter 25).

Bar swing, release and catch

In the bar swing, a classical movement from gymnastics, the momentum provides an additional

mechanical stimulus for the connective tissues. For the ninja, it is the basic stage before progression to other bar exercises. From the swing it is possible to release the bar and to catch another object (see Figure 35.6). This technique takes time to learn, not only to find optimal timing, moreover to prepare the upper extremities for the high eccentric load and deceleration skill that puts tremendous load on the tissues. It is advisable that training volume and frequency are chosen carefully to ensure sufficient recovery for

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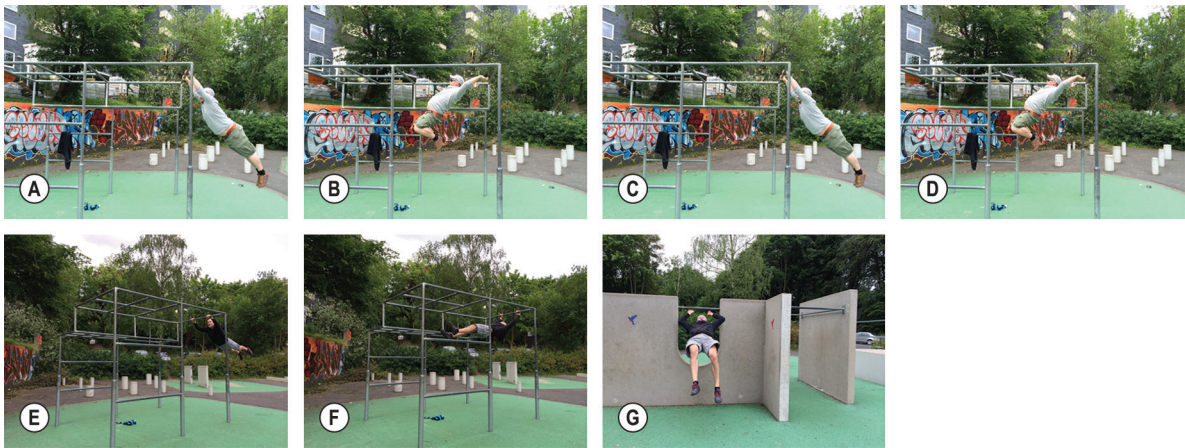


Figure 35.6

(A, B) Flying from bar to bar is a skill that even overweight people can master. (C, D) The difference to more advanced individuals is movement dynamics and the range of movement due to a superior strength-bodyweight ratio. (E, F) Using an underwing can be used in a breakthrough situation (G).

injury prevention. In the beginning you should focus on learning to control the deceleration in the eccentric phase. This can be achieved by eccentric exercises like lowering the body from the bar. Start with small jump distances of 10 to 20 cm, then gradually increase the distance. Mastering this skill requires lots of grip and shoulder strength, a clear example that you need strong muscles first to exercise fascia connective tissues effectively.

Supporting movements

In order to overcome obstacles supporting movements are fundamental and they are extremely rich in variations. Figure 35.7 illustrates a creative variant of the monkey or kong vault. We can observe three supporting contacts on the table that lead to a 180 degree turn of the body and a backwards landing. After the jump

off the supporting contacts start in the first third of the table leading to short walking-style hand movements.

Summary

The inductive approach in this chapter demonstrates that exercise and the fitness of fascia can take many forms. Although there are no scientific studies elucidating its effectivity in this regard, ninja style or parkour training contains many components that may be able to trigger significant adaptations in the fascial connective tissue. In this sense, the high scalability and variability could make parkour movements a true fascial fitness exercise mode. Even although these urban subcultures seem to appeal more to teenagers to date, they could be a useful endeavor for adults wanting to break out of their usual movement patterns and habits.

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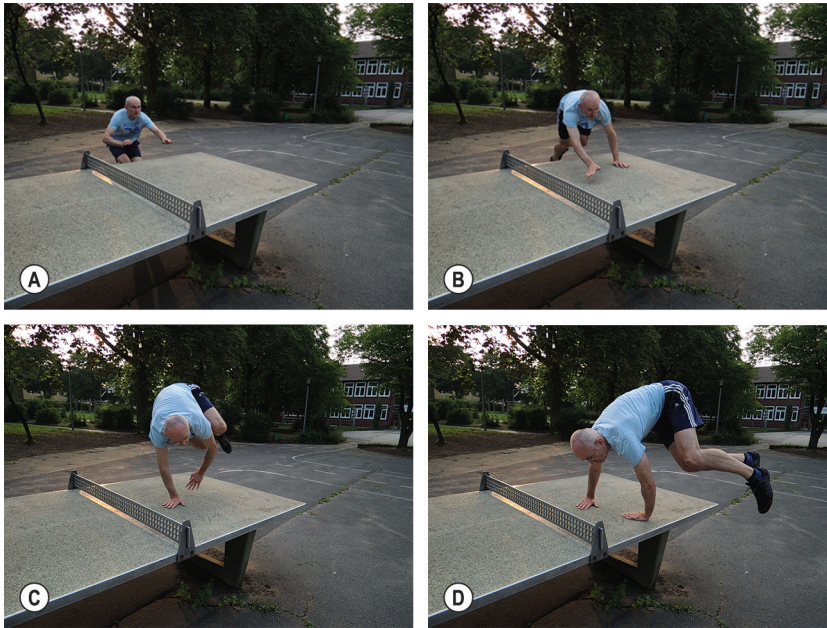


Figure 35.7

(A–D) Creative variation of an obstacle jump with three supporting hand contacts on a ping-pong table.

References

Findley T.W, Schleip R. (eds) (2007) *Fascia Research: Basic Science and Implications for Conventional and Complementary Health Care*. Munich, Germany: Elsevier Urban & Fischer.

Heinemann, K. (2007) *Introduction to the Sociology of Sport: Basics for Studies, Training and Work*. Schorndorf, Germany: Hofmann-Verlag GmbH & Co.

Hirtz, P. (2003) Koordinationstraining. In: Schnabel G., Harre, D., Krug, J., *Trainingswissenschaft: Leistung – Training – Wettkampf*. Germany: Meyer + Meyer Fachverlag.

Kläber, M. (2014) *Modern Muscle Cult: On the Social History of Bodybuilding*. Bielefeld, Germany: transcript Verlag.

Luksch, M. (2009) *Tracers Blackbook: Techniken im Parkour Training*. Germany: ParkourONE.

Sano, K., Kunimasa, Y., Oda, T., Nicol, C., Komi, P.V., Locatelli, E., Ito, A., Ishikawa, M. (2014) Achilles tendon moment arm for Kenyan runners. *Scand J Med Sci Sports*. 24: e269–e274.

Scheid, V., Prohl, R. (2012) *Sport and Society*. 7. Wiebelsheim, Germany: korrigierte Auflage.